

SOLUTIONS

Module:	Fundamentals of DSP		
Module Code	QHE5107	Paper	B
Time allowed	3 hrs	Filename	"QHE5107-Fundamentals of DSP - Resit - Draft version - Without Answers"
Rubric	ANSWER ALL FOUR QUESTIONS		
Examiners	Dr Fabrício de Oliveira Ourique	Dr Jesús Requena Carrion	



北京邮电大学



Queen Mary
University of London

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QHE5107 B

Queen Mary School Hainan 2024/25

QHE5107 Fundamentals of DSP

Paper B

Time allowed 3 hours

Answer ALL questions

Complete the information below yourself very carefully.

QM student number

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BUPT student number

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Class number

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NOT allowed: Any Electronic devices including calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must not take answer books, used or unused, from the examination room.**
2. Write only in black or blue pen **and in English.**
3. Do all rough work in the answer book – **do not tear out any pages**
4. If you use Supplementary Answer Book, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Fabrício de Oliveira Ourique, Dr Jesús Requena Carrion

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Filename: "QHE5107-Fundamentals of DSP - Resit - Draft version - Without Answers"
required

No answerbook

Question 1**[25 marks]**

Consider the following difference equation of a causal system:

$$y[n] = 3x[n] + \frac{1}{4}y[n-1] + \frac{1}{8}y[n-2]$$

(a) Obtain the system function, $H(z)$, and the impulse response, $h[n]$. Determine whether the system is BIBO stable. **(5 marks)**

Solution:

The system function is

$$H(z) = \frac{3}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}} \quad (\text{Basic: 1 mark})$$

The impulse response is

$$h[n] = 2(0.5)^n u[n] + (-0.25)^n u[n] \quad (\text{Advanced: 2 marks})$$

To be stable, we need

$$\sum_{n=-\infty}^{\infty} |h[n]| < \infty \quad (\text{Basic: 1 mark})$$

$$\sum_{n=-\infty}^{\infty} |h[n]| = 4.8$$

The system is stable. (Basic: 1 mark)

(b) Is the system recursive? **(5 marks)**

Solution:

Since $y[n]$ depends on $\frac{1}{4}y[n-1] + \frac{1}{8}y[n-2]$, the system is recursive, (Basic: 3 marks)
 as it uses past output values in determining the current output. (Basic: 2 marks)

(c) Is the system FIR? **(5 marks)**

Solution:

The impulse response is

$$h[n] = 2(0.5)^n u[n] + (-0.25)^n u[n] \quad (1)$$

The impulse response $h[n]$ of an FIR system is finite, meaning it has non-zero values only for a limited number of samples and becomes zero afterward. (Basic: 3 marks)

Therefore, the system is NOT FIR. (Medium: 2 marks)

(d) Find the system output when $x[n] = \left(-\frac{1}{2}\right)^n u[n]$. (10 marks)

Solution:

$$Y(z) = X(z)H(z) \quad (\text{Basic: 2 marks})$$

$$X(z) = \frac{1}{1 + 0.5z^{-1}} \quad (\text{Medium: 2 marks})$$

$$H(z) = \frac{3}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}} \quad (\text{Advanced: 1 mark})$$

$$Y(z) = \frac{1}{1 + 0.5z^{-1}} \frac{3}{1 - \frac{1}{4}z^{-1} - \frac{1}{8}z^{-2}} \quad (\text{Advanced: 2 marks})$$

$$y[n] = (0.5)^n u[n] + 3(-0.5)^n u[n] - (-0.25)^n u[n] \quad (\text{Medium: 3 marks})$$

Question 2

[25 marks]

Consider a system that multiplies a given input by a ramp function, $r[n]$. That is,

$$y[n] = x[n]r[n] = x[n]nu[n].$$

(a) Describe how the ramp function $r[n]$ might affect the frequency response of the system. Specifically, discuss whether the multiplication by n impacts specific frequency components of $x[n]$. (5 marks)

Solution:

To understand the frequency response, observe that multiplying $x[n]$ by n (the ramp function) in the time domain results in a different effect in the frequency domain. (Advanced: 3 marks)

Multiplying a signal by n in the time domain introduces a differentiation effect in the frequency domain, which changes the spectrum of $x[n]$. Specifically, the ramp function increases the weight of higher-frequency components, leading to distortion in the original signal. (Medium: 2 marks)

(b) Is the system linear? (5 marks)

Solution:

Given

$$y[n] = x[n]nu[n]$$

For two inputs $x_1[n]$ and $x_2[n]$, the output are:

$$y_1[n] = x_1[n]nu[n]$$

$$y_2[n] = x_2[n]nu[n]$$

If we apply an input $ax_1[n] + bx_2[n]$, the output is

$$y[n] = (ax_1[n] + bx_2[n])nu[n] \quad (\text{Medium: 2 marks})$$

$$y[n] = ax_1[n]nu[n] + bx_2[n]nu[n]$$

$$y[n] = ay_1[n] + by_2[n] \quad (\text{Basic: 2 marks})$$

The system is linear.

(Basic: 1 mark)

(c) Is the system memoryless?

(5 marks)

Solution:

A system is memoryless if the output at any time n depends only on the input at that same time n , without relying on past or future input values. (Basic: 2 marks)

For the system:

$$y[n] = x[n]nu[n]$$

the output $y[n]$ at time n depends on $x[n]$ and the value $nu[n]$. (Advanced: 2 marks)

Therefore, the system is memoryless (Basic: 1 mark)

(d) Is the system causal?

(5 marks)

Solution:

A system is causal if the output $y[n]$ at any time n depends only on the current or past values of the input $x[n]$, not on future values. (Medium: 1 marks)

In this case, $y[n] = x[n]nu[n]$ depends only on $x[n]$ at the current time n and does not require future values of $x[n]$. (Medium: 2 marks)

Therefore, the system is causal. (Basic: 2 marks)

(e) Is the system time-invariant?

(5 marks)

Solution:

For an input $x[n]$,

$$y[n] = x[n]nu[n]$$

For an input $x[n - k]$,

$$y_k[n] = x[n - k]nu[n]$$

We know that

$$y[n - k] = xn - ku[n - k] \quad (\text{Basic: 2 marks})$$

As

$$y_k[n] \neq y[n - k] \quad (\text{Medium: 2 marks})$$

The system is not time invariant. (Basic: 1 mark)

Question 3**[25 marks]**

A compact disc (CD) records audio signals digitally by means of a binary code. Assume an audio signal bandwidth of 15 kHz , i.e. $X_a(f) = 0, |f| > 15\text{ kHz}$.

(a) What is the minimum sampling frequency to avoid aliasing distortion? **(5 marks)**

Solution:

$$f_S > 2f_{max} \quad (\text{Medium: 2 marks})$$

$$f_S = 30\text{kHz} \quad (\text{Basic: 3 marks})$$

(b) If the samples are quantized into 65,536 levels ($L = 65,536$) and then binary coded, what number of binary digits is required to encode a sample. **(5 marks)**

Solution:

$$\text{Bits} = \log_2(65,536) \quad (\text{Medium: 3 marks})$$

$$\text{Bits} = 16 \quad (\text{Basic: 2 marks})$$

(c) Determine the number of binary digits per second (bits/s) required to encode the audio signal. **(5 marks)**

Solution:

$$\text{number of bit per second} = 30,000 \times 16 \quad (\text{Advanced: 3 marks})$$

$$\text{number of bit per second} = 480,000\text{bit/s} \quad (\text{Basic: 2 marks})$$

(d) For practicality, signals are sampled at a rate well above the Nyquist rate. Practical CDs use a sample rate of 44,100 samples per second. Given that $L = 65,536$, determine the number of bits per second required to encode the signal per channel and for two audio channels (stereo). If a standard CD can store up to 700MB of data, calculate the approximate maximum duration (in minutes) of CD-quality stereo audio that can be stored. **(10 marks)**

Solution:

$$\text{number of bit per second} = 44,100 \times 16 \quad (\text{Basic: 3 marks})$$

$$\text{number of bit per second} = 705,600 \text{bits/s per channel} \quad (\text{Medium: 2 marks})$$

$$\text{number of bit per second} = 2 \times 705,600 \text{bits/s Stereo} \quad (\text{Medium: 2 marks})$$

$$\text{number of minutes} = \frac{700MB}{2 \times 705,600 \text{bits/s}} \frac{1}{60 \text{s/min}} \approx 66 \text{min} \quad (\text{Advanced: 3 marks})$$

Question 4**[25 marks]**

Consider the following desired impulse response for a second-order FIR filter:

$$h[n] = \delta[n] - \delta[n - 2]$$

(a) Write the difference equation for this FIR filter.

(5 marks)**Solution:**

$$y[n] = x[n] \star h[n] \quad (\text{Basic: 1 mark})$$

$$y[n] = x[n] \star (\delta[n] - \delta[n - 2]) \quad (\text{Basic: 2 marks})$$

$$y[n] = x[n] - x[n - 2] \quad (\text{Advanced: 2 marks})$$

(b) Calculate the frequency response, $H(e^{j\omega})$, of the filter.

(5 marks)**Solution:**

$$H(e^{j\omega}) = \mathcal{F}\{h[n]\} \quad (\text{Basic: 1 mark})$$

$$H(e^{j\omega}) = \mathcal{F}\{\delta[n]\} - \mathcal{F}\{\delta[n - 2]\} \quad (\text{Basic: 2 marks})$$

$$H(e^{j\omega}) = 1 - e^{-j2\omega} \quad (\text{Medium: 2 marks})$$

(c) Find $|H(e^{j\omega})|$ and plot the magnitude response of the filter for frequencies between $-\pi$ and π . Describe the type of filter (e.g., low-pass, high-pass, etc.) based on the plot. **(10 marks)**

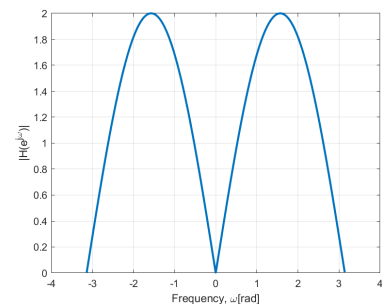
Solution:

$$H(z) = 1 - z^{-2} = (1 - z^{-1})(1 + z^{-1}) \quad (\text{Medium: 2 marks})$$

$$H(e^{j\omega}) = (1 - e^{-j\omega})(1 + e^{-j\omega}) \quad (\text{Basic: 2 marks})$$

$$|H(e^{j\omega})| = |1 - e^{-j\omega}| \cdot |1 + e^{-j\omega}| \quad (\text{Basic: 2 marks})$$

$$|H(e^{j\omega})| = |e^{j\omega} - 1| \cdot |e^{j\omega} - (-1)| \quad (\text{Advanced: 2 marks})$$



Band-Pass Filter (Advanced: 2 marks)

(d) Explain how this FIR filter would affect a discrete input signal containing frequency components at 0 and π radians. **(5 marks)**

Solution:

At $\omega = 0$ (DC frequency)

$$H(e^{j0}) = 1 - e^{-j0} = 1 - 1 = 0 \quad (\text{Basic: 1 mark})$$

The frequency response at $\omega = 0$ is 0, meaning that any DC component of the signal is completely attenuated by the filter. (Medium: 1 mark)

At $\omega = \pi$

$$H(e^{j\pi}) = 1 - e^{-j2\pi} = 1 - 1 = 0 \quad (\text{Basic 1 mark})$$

The frequency response at $\omega = \pi$ is also 0, meaning that any frequency component at π radians (the Nyquist frequency) is also completely attenuated by the filter. (Medium: 1 mark)

Since we have two zeros at the frequencies 0 and π radians, the signal components at these frequencies will be blocked or filtered out. (Medium: 1 mark)